



Invention Article

Explainable hybrid AI for job recommendation

Snehadhanya Dutta ^a, Moumita Chatterjee ^b, Dhrubasish Sarkar ^{c,*} ^a Srikrishna College, Bagula, Nadia, West Bengal, India^b Aliah University, Kolkata, West Bengal, India^c Bhawanipur Global Campus, Kolkata, West Bengal, India

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ABSTRACT

The growing popularity of automated processes, empowered by Artificial Intelligence (AI), is changing how companies offer and fill their jobs. The growth of AI has created the need for a more sophisticated way to find a job, through the use of CRM-based applications. Most job search sites offer simple keyword matching (keyword searches) with little or no customization, resulting in a low accuracy rate. This paper presents a new type of Job Recommendation Model, based on the use of Hybrid AI models. The purpose of this new hybrid model is to improve job matching through three separate processes: skill-based filtering, sentiment analysis, and predictive modeling. The proposed hybrid model will take user calculated skill sets into account for the job description and will use AI adoption levels, automation risk, expected salary, and anticipated growth opportunity for the job type. The hybrid model will produce two scores for each job type; the first score represents the potential to match with the specific job type based on the applicant's skills and the second score represent the applicants' overall profile. Once combined to create a matching score, the job recommendation system will recommend jobs based upon applicant's (1) Skills, (2) Ability to do the job, (3) Salary Range, and (4) Job Growth Potential (area of interest on the applicant).

1. Specifications table

Subject code	1702
Specific subject area	Hybrid AI-based job recommendation systems using machine learning, NLP, and explainable AI for skill-based career matching.
Industry code	Y/Y04/Y04S
Details of inventors	
Dates of invention	Conception: 15 September 2024 Disclosure: 5 March 2025
Patent details (only if patented)	NA
Intended use	The software-based invention is intended for research use, collaborative development, and potential licensing, with the possibility of future partnership or commercialization following further validation and development.

(continued)

	https://doi.org/10.14419/pr5w2v12
	Hadjer khider, Abdelkrim Meziane, and Slimane Hammoudi. 2025. Toward an Approach for Job Recommender System: Leveraging Hybrid Techniques. In Proceedings of the 2024 8th International Conference on Natural Language Processing and Information Retrieval (NLPPIR '24). Association for Computing Machinery, New York, NY, USA, 304–309. https://doi.org/10.1145/3711154.2.3711608
Software availability	NA
Related other sources: (datasets, software, diagrams, plans, etc.)	NA

Related research article	Soni, S., & Shah, S. (2025). Job Recommendation Systems Through AI and Machine Learning. International Journal of <i>(continued on next column)</i>
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2. Value of the software

- The system developed improves the job recommendation process by overcoming fundamental downfalls of current keyword-based

* Corresponding author.

E-mail address: dhrubasish@inbox.com (D. Sarkar).

platforms. This will allow for the generation of job recommendations that are indicative of both an individual's job fit and the larger scale of employment trends; therefore, resulting in more reliable and contextually based career counselling.

- The model being developed is beneficial to job seekers looking for career paths that are aligned with their skills, recruitment websites and companies looking to improve their recommendations, HR Analytics Analysts who are assisting with hiring, as well as researchers who are studying A.I. employed by the labor market and incoming workforce trends.
- The architecture will serve as an example of the potential of hybrid recommendation system frameworks and can be repurposed for ongoing research into identifying skill shortages, workforce forecasting, and similar recommendations relating to education, training and skill development.
- The inclusion of Explainable A.I. legally requires robotic and automated systems including that supporting employment-related decision-making to provide insight into the factors that influenced their decisions including: skill sets, salary aspirations, and other job specifications. This provides for accountability, transparency, and responsible use of robotic systems for employment-based decision-making.
- The interchangeable modular design allows for the integration of new labor market information as it becomes available, new skill information, and new learning systems, to remain relevant and beneficial as new labor practices and skill demands evolve.

3. Software based invention description

3.1. Overview of the software

A new hybrid AI-based job recommendation system is being developed, which generates personalized job recommendations through the combination of several different techniques. These include Skill Matching, Sentiment Analysis, Labor Market Attributes, Predictive

Machine Learning Models, and Explainable AI Techniques. Both user profile data and job listing data are processed using a single unified pipeline in order to calculate the job suitability score for each job and generate a ranking of jobs by the suitability score.

3.2. System architecture and processing flow

The overall architecture of the software follows a modular pipeline in which data ingestion, preprocessing, feature extraction, filtering, predictive modelling, explainability, and output generation are executed sequentially, as illustrated in the hybrid framework diagram (Fig. 1) included in the paper. User and job data are first converted into structured representations and progressively refined, ensuring that only relevant job candidates proceed to the predictive stages.

3.3. Input data and representation

Two types of data are required for candidate selection from the database: User Profiles and Job Listings. The User Profile has data such as Skills, Years' Experience, Job Type and Salary Expectations. The Job Listing has data such as Job Title, Job Description, Required Skills, Salary Range, Likelihood of AI Adoption, Risk of Automation for that position, and Projected Growth for that position. These two pieces of information can be represented in both Numeric and Text Formats that can be used with machine learning algorithms to assist in determining which candidates are the best match for more than one person or career or position.

Data used in this study contains approximately 50,000 records (includes both user profiles and job listings). The data was acquired from different publicly available employment websites (LinkedIn, Indeed, Glassdoor, Monster, Naukri) to create a comprehensive representation of the job market across many different professional industries and job positions.

The dataset being used in this study contains important pieces of information, the major ones being: User Skills, User Experience, Salary

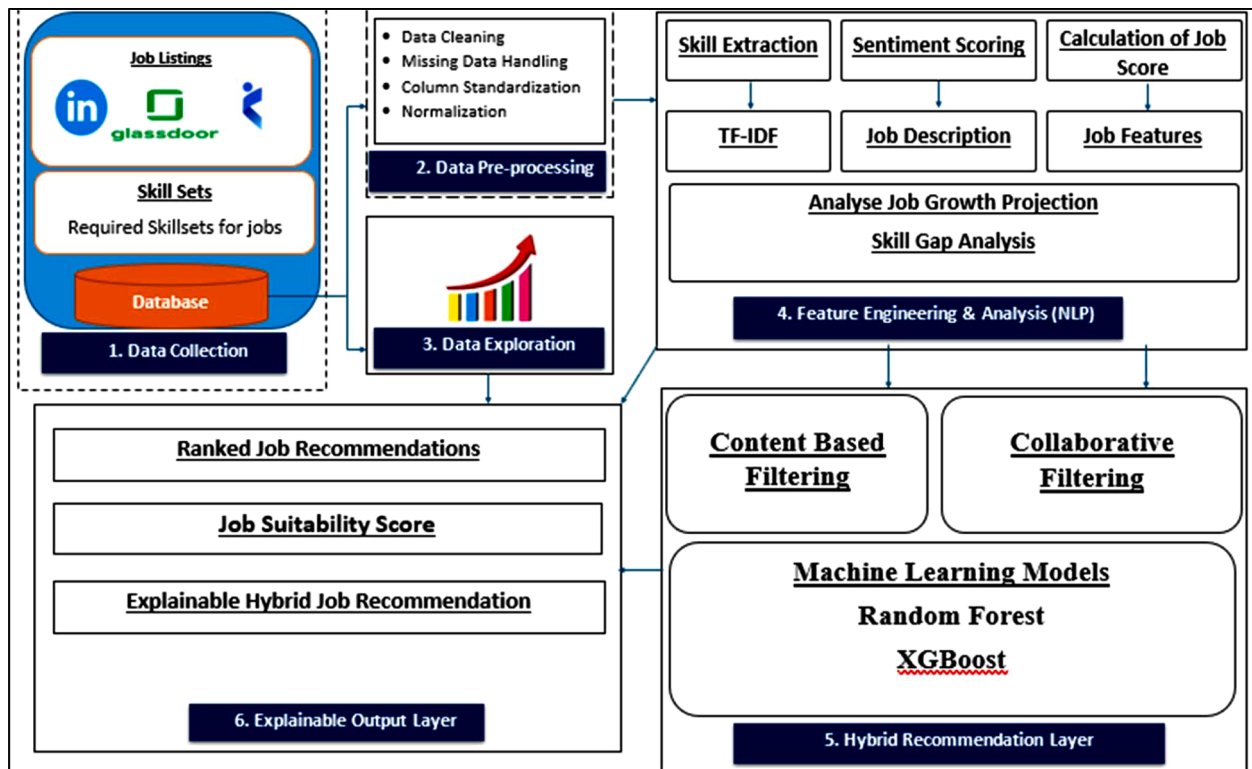


Fig. 1. Proposed model framework.

Expectations, Job Description, Required Skills, Salary Range, Projected Job Growth, and AI Adoption likelihood, and Automation Risk, which together create a multi-source and varied dataset will allow the model to establish actual job market patterns and better model the relationship between User Profiles to characteristics of Job Listings.

3.4. Data preprocessing

In order for accurate, standardized, and comparable analysis to occur before models are built, the dataset will undergo preprocessing that consists of multiple steps; i.e., deleting duplicate records, handling missing data, converting categorical variables into useable formats, standardizing numerical data. All preprocessing methods will help maintain consistency of data and enhance overall performance of model.

Textual data such as job descriptions and skills will be cleaned and tokenized so that they are ready for use in natural language processing (NLP) applications. After NLP processes are completed, text data will have a structured format that allows for filtering prior to being passed into machine learning models.

The dataset will be split into two parts (80%/20%) for training/testing purposes in order to allow the model to use enough data to establish a correlation between predicted responses and actual job performance. Since continuous job suitability scores will be predicted rather than discrete categorical labels there is no meaningful class imbalance, therefore additional class balancing is not going to be needed.

While splitting the data into two parts creates a consistent methodology/standard for model validation, future work should use k-fold cross-validation techniques to further improve the quality of prediction results produced by model.

3.5. Feature extraction and skill representation

Methods listed in Table 1 are used to generate a list of skills and preferences. Job descriptions were analyzed using TF-IDF to find the relevant skills associated with the position; subsequently, the cosine similarity was employed to establish a match between the skills the user has and the skills required for the job. A quantitative measure of the match of the skill sets (to each other) is thus produced.

NLP techniques are utilized to conduct a sentiment analysis on the postings, producing sentiment scores based on the overall tone implied by each posting. These scores could assist with capturing the appeal of job postings to users based on their perceptions and/or preferences.

Finally, additional measures of the user (e.g., expected pay, desired experience, etc.) are quantified and averaged. The skills identified previously, the sentiment score associated with each posting and the user profile attributes are combined into a common feature vector for each job-user combination, allowing for the more relevant overall job recommendations.

3.6. Filtering mechanisms

Initial job screening is performed through a combination of content-based and collaborative filtering, following the steps outlined in Table 2. Content-based filtering evaluates direct compatibility between user

attributes and job requirements, focusing on skill similarity, salary alignment, and experience level. Collaborative filtering then incorporates information from users with similar profiles to refine recommendations and expose users to relevant roles beyond direct matches.

These two approaches are combined in a hybrid filtering stage, balancing individual preferences with collective patterns to generate a refined set of candidate jobs for predictive evaluation.

3.7. Predictive modelling

Filtered applicants for job openings are assessed for job suitability with machine-learning models to produce a suitability score. XGBoost uses predictive scoring to account for non-linear relationships between characteristics (skills, salary alignment, AI adoption rate, and job growth projection), which is shown in the prediction representations of salaries and job scores in Fig. 2. Random Forest uses an aggregate approach to combine predictions from several decision trees to provide a strong estimate of features that relate to sentiment impact and automation risk, shown in the feature importance visualization (Fig. 3).

The complete predictive workflow, including model configuration parameters and feature contributions, is summarized in the predictive modelling table (Table 3).

3.8. Model evaluation

The prediction accuracy and stability of an output are evaluated using evaluation metrics systematically and uniformly applied for all job types this includes Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE), these error values quantify how far a predicted job suitability score is from the actual job suitability score and the evaluation plots will visually present these values (see Table 4). The use of RMSE and MAPE provides an appropriate measure for all continuous prediction tasks providing uniformity among different job types of all job attribute scales.

In the absence of explicit user interaction data (such as clicks or applications), true ground truth suitability is not directly available. As a result, evaluation focuses on the consistency and stability of predicted suitability scores across different job categories. While this enables systematic assessment, it represents an approximation rather than validation against real user-driven relevance.

The current research integrated three methodologies i.e. content-based filtering, collaborative filtering, predictive modelling in an overall system designed as an integrated framework rather than three separate systems that work independently. An integrated system has the ability to improve performance compared to the individual systems.

To further validate and support the robustness or generalizability of these results, explicit baseline comparisons can be provided and evaluated, as well as formal tests for statistical significance.

3.9. Explainable AI integration

SHAP is used in this system to provide an explanation at the feature-level for every prediction. Each feature's contribution to the predicted values are computed and shown on SHAP summary plots (Fig. 4). SHAP

Table 1

Feature engineering components and their role in job recommendation accuracy.

Feature Type	Methodology	Purpose	Contribution to Accuracy
Skills	TF-IDF & Cosine Similarity	Identifies relevant skills from job descriptions	Enhances alignment between user skills and job requirements, improving personalized recommendations.
Sentiment	Sentiment Analysis	Analyzes job descriptions to determine tone	Helps categorize jobs as positive, negative, or neutral, improving the relevance of job matches based on user sentiment preferences.
User Preferences	NLP-based Personality Analysis	Infers traits from user-entered data	Aligns job recommendations with user personality, enhancing suitability and user satisfaction.

Table 2
Steps involved in extracting skills, sentiment for job recommendations.

Component	Description	Purpose	Contribution to Recommendation
Content-Based Filtering	Matches user-entered skills with job requirements, considering factors like salary and experience levels.	Aligns job recommendations with user-specified skills and criteria.	Enhances personalization by focusing on user-specific attributes.
Collaborative Filtering	Identifies users with similar skills, salary ranges, and experience (e.g., 80% match or higher).	Leverages preferences of similar users to refine recommendations.	Improves relevance by considering broader trends among similar users.
Machine Learning-Based Predictions	Uses models like XGBoost and Random Forest to generate job scores (range: 0–100) based on features.	Predicts job suitability using factors like skills (30%), salary (20%), and AI adoption (10%).	Offers precise recommendations by incorporating predictive analysis.
Hybrid Recommendation System	Combines content-based and collaborative filtering approaches to balance user-specific (50%) and collective behaviors (50%).	Integrates individual preferences with shared user trends.	Increases accuracy and relevance by combining personal and shared attributes.

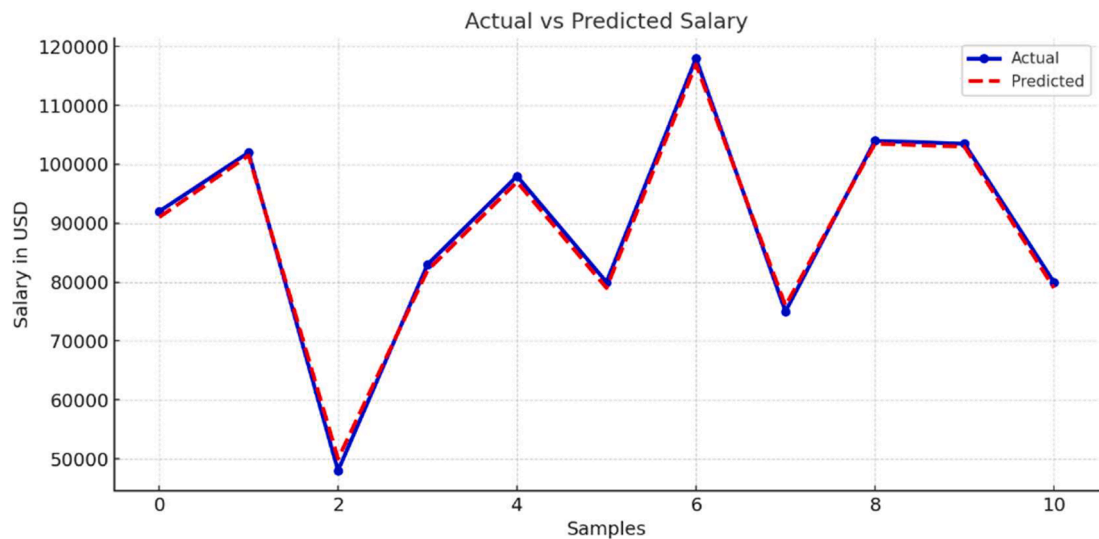


Fig. 2. Predicting salary score – XG boost.

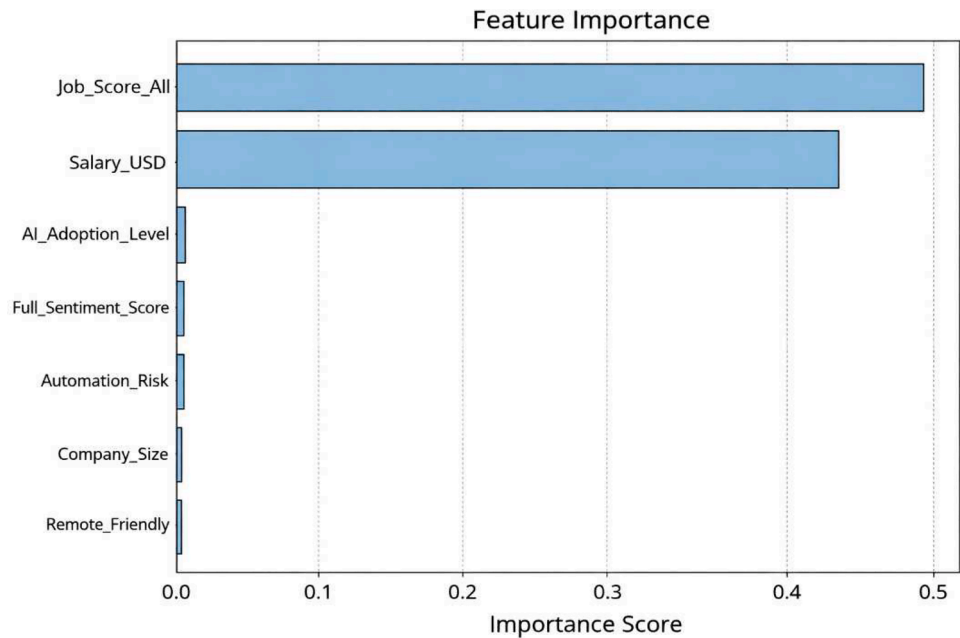


Fig. 3. Predicting sentiment score by feature importance – random forest.

summary plots display how job suitability score is affected by certain features like skills, salary expectations, and job growth.

For each recommendation given by the system, SHAP values identify how much each feature contributes positively or negatively to the

predicted value of the recommendation and provide an easy way to see how the overall prediction was made. Thus, users will have a clear view of how the SHAP values will help understand the behavior of the model and how each recommendation was arrived at.

Table 3

. Steps in the predictive modelling workflow to the prediction.

Step	Description	Purpose	Contribution to Job Suitability Prediction
XGBoost	Optimized gradient boosting model, uses up to 100 iterations.	Captures complex, non-linear relationships.	Improves accuracy by handling interactions among up to 50 features.
Random Forest	Ensemble model with 100–200 decision trees.	Reduces variance, enhances stability.	Provides robust predictions by averaging across multiple trees.
Explainable AI (XAI)	Uses SHAP values to interpret feature contributions (0–1 scale).	Ensures transparency & interpretability.	Shows the impact of each feature on prediction, with SHAP values indicating importance (e.g., skill contribution: 0.3).

Transparency is especially critical in job recommendation systems because it permits users of the system to see what information was used for each of the recommendations. The user can also indicate which part of the input contributed to their recommendation; therefore, helping them to make more informed and confident choices.

By integrating explainable AI into the model, the model can support the principles of human centered design by outputting interpretable results in employment-related areas where it can be damaging to make automated decisions about people's employment without providing an explanation.

3.10. Output generation and classification

The output generation stage consolidates results from the filtering,

predictive modeling, sentiment analysis, and explainability components of the system. For each user job pair that passes the hybrid filtering stage, the system generates a set of quantitative outputs that determine the ranking and classification of job recommendations, consistent with the experimental evaluation.

In determining a user's job suitability score, the score is computed through the use of weighted combinations of features, where the weightings are determined quantitatively; hence, it is possible to measure and compare the relative significance of the various features, as the weights allotted to each of the features represent its respective importance for determining relevance to a job. For example, skills constitute the highest weighted feature because they directly relate to a user's compatibility with the job. Job title and salary attributes also receive significant weightings due to how they align with preferences and practicality of use for job matching.

Other features that are assigned moderate weightings (AI adoption risk, automation risk and job growth projection) are included so that the recommendation algorithm includes an evaluation of outside macro-level trends that affect the overall job market, but not to the extent that they would overshadow the core components used in the importance of matching to a job.

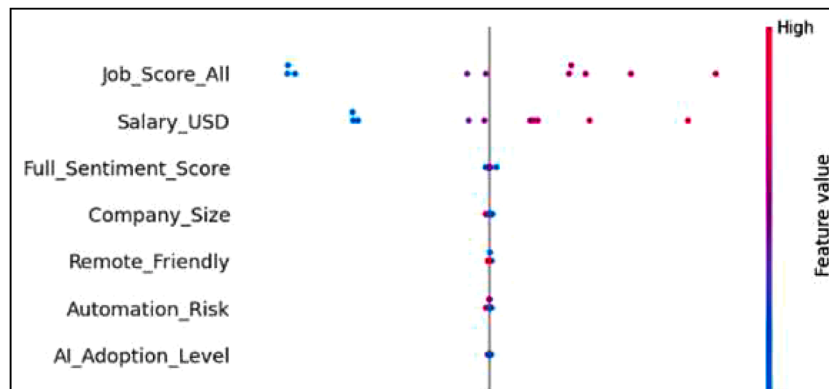
The overall weightings used to compute a user's job suitability score (as well as the various input features used to generate the output) are intended to help balance the influence of information at the user specific level with macro-level trends associated with job markets. As they are defined at this time, rather than with quantified data-based methods, the future use of quantified data and weighted forecasting through machine learning techniques could further lead to ongoing optimization of the weightings currently being used to compute a user's job suitability score.

At the core of this stage is the job suitability score, which is calculated using the weighted feature structure defined in Table 5. Job title relevance, skill match ratio, salary alignment, AI adoption level,

Table 4

Values of RMSE & MAPE – random forest.

Job	Random Forest							
	Salary		Job Growth		Job Score		Sentiment	
	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
Cybersecurity Analyst	1559.3	0.02	0.33	5.79	305.15	0.02	0	0
Marketing Specialist	1284.9	0.01	0.54	1.31	256.17	0.01	0.01	0.02
AI Researcher	820.53	0.01	0.51	7.36	147.26	0.01	0	0.01
Sales Manager	869.44	0.01	0.59	6.48	179.96	0.01	0.01	0.01
UX Designer	5037.4	0.03	0.42	6.14	1012.42	0.03	0.01	0.01
HR Manager	1364.6	0.01	0.47	3.79	290.32	0.02	0.01	0.01
Product Manager	3952	0.03	0.32	3.2	584.17	0.02	0.01	0.02
Software Engineer	9235.7	0.03	0.48	9.25	1859.13	0.03	0.01	0.02
Data Scientist	780.58	0.01	0.46	6.85	189.25	0.01	0.01	0.01
Operations Manager	3812.1	0.02	0.44	9.5	755.07	0.02	0.02	0.03

**Fig. 4.** SHAP summary plot for hybrid recommendations.

automation risk, and job growth projection each contribute to this score according to predefined weights. This ensures that the scoring logic used during experimental analysis is preserved during operational output generation.

Alongside suitability scoring, a sentiment score is generated for each job through NLP-based analysis of job description text. These scores capture the overall tone of job postings and are retained as independent outputs. Average sentiment values across job titles, reflected in Table 6, enable comparison of qualitative job characteristics across different roles and industries.

To assess how well the candidate's skills match that of a specific job, we use cosine similarity to compare the candidate's declared skill set with the requirements of the job. The skill match score is the main basis for determining whether or not a candidate will be successful in their application. In both instances, the skill match score will provide information regarding the extent to which a candidate's skill set aligns with those required for a specific job, using the evaluations of the feature sets in Table 7. The feature combination that produces the highest overall score is the combination of skill match and job growth projection.

In addition, the system generates a predicted score using machine-learning models. This score represents the final output of the XGBoost and Random Forest predictions and captures non-linear interactions among features such as skills, salary alignment, AI adoption level, and automation risk. Model behavior and stability are validated using the predictive workflow outlined in Table 3.

In addition to providing individual recommendation scores, the software generates aggregated data by job role from the primary dataset. This aggregation includes information relating to Company Size Average, Average AI Adoption Rate, Average Automation Risk, Average Salary Range, Remote Work Availability, and Job Growth Rates by Title for each Job Title. Table 8 shows the Role Level Characteristics of each Job Title and their corresponding Role Level Recommendation Scores. These Characteristics provide context for the Role Level Recommendation Outputs, and therefore ensure that each Role Level Recommendation Score is generated in accordance with the labor market trends observed.

Using the computed job suitability and sentiment scores, recommendations are categorized into high, moderate, and low suitability classes based on predefined numerical thresholds specified in Table 9. These thresholds are applied consistently during output generation to maintain alignment between experimental classification and operational system behavior.

When the recommendations are complete for an individual, they contain a list of the top jobs for that individual (Table 10). Each job on the job recommended list will include its title, industry, salary, job growth indicator (indicator of how many jobs will be created), sentiment score (rating of how happy people are working at that job), job suitability score (rating of how well an individual's skill set fits that position), predicted score (predicted score based upon the above variables) and skill match score (determination of how closely the person's skills

Table 5
Initial job score calculation metrics.

Feature	Weight (%)	Matching Criteria	Description
Job Title	20	Matching job designation	Compares user's preferred job title with the job listing.
Skills	30	Skill match ratio	Ratio of matching user-inputted skills to job skills.
AI Adoption	10	AI-centric job indicator	Determines AI influence in the job role.
Automation Risk	10	Automation likelihood	Measures potential for job automation.
Salary	20	Salary alignment	Compares user salary expectations with job offer.
Job Growth Projection	10	Future job stability	Predicts growth trends for job roles.

Table 6
Average job score and sentiment score of the unique job titles.

Job Title	Job Score	Sentiment Score
Cybersecurity Analyst	22,278.788325	0.526384
Marketing Specialist	18,758.724782	0.311443
AI Researcher	21,434.370919	0.341525
Sales Manager	18,605.647418	0.283334
UX Designer	20,565.301398	0.249124
HR Manager	20,413.296605	0.262353
Product Manager	20,200.825049	0.224107
Software Engineer	15,680.697144	0.226438
Data Scientist	16,359.199900	0.378221
Operations Manager	17,708.219635	0.267770

Table 7
Combined job and sentiment scores.

Feature Combination	Job Score (%)	Sentiment Score (%)	Overall Score (%)
Skills + Job Growth Projection	85	80	83
AI Adoption Level + Automation Risk	75	70	73
Salary USD + Job Title	80	78	79

match the requirements of the job). This structured format of output matches the format of the results for the final Hybrid Recommendation Algorithm (de Montemayor, 2005), and it illustrates how experimental evaluations resulted in the final user-facing Recommendation List. All values produced as outputs, classifications produced, and final materials produced as outputs via the Hybrid Recommendation Algorithm contain the same definitions, weights, and thresholds as those used in the experimental evaluations and therefore will produce the same outcome irrespective of when or where the software is executed.

3.11. Score normalization

Feature score differences, such as skill match, sentiment score, and predicted suitability, may exist across various numerical values/scales. All feature scores should first be **normalized** onto a single scale prior to aggregation to ensure they all contribute equally and consistently to the overall final job suitability score. Additionally, this helps to provide balance between the aggregates of each of the feature scores and to provide a greater degree of reliability with respect to the final job suitability score.

3.12. Characteristics and utility of the software

The module-based prediction engine processes both structured and unstructured data using a single unified process to filter and generate models as well as provide explanations regarding all the predictions. This architecture, flexibility allows it to be used in platforms that provide job recommendations, job matching, HR analytics platforms, career guidance systems where transparent recommendations are necessary.

Also, the recommendation quality will ultimately improve by having multiple factors considered; such as matching skills of a candidate with job growth, automation risk and so on. This ultimately leads to more accurately matching jobs to the candidates based on these multiple dimensions and provides more contextual appropriation for job matches.

The addition of Explainable AI will further enhance transparency and trust in the recommendations through interpretability. Therefore, users will receive job matching recommendations and see how the individual outputs they received occur based on their various features, leading to more reliable and usable products overall.

Table 8

Summary table outlining the average characteristics of the job roles.

Job Title/Avg.	Company Size	AI Adoption Level	Automation Risk	Salary (\$)	Remote Friendly	Job Growth	Job Count
Data Scientist	50.8%	49.2%	43.5%	91,481	54.8%	49.2%	12.4%
HR Manager	52.6%	46.5%	49.1%	89,252	59.6%	42.9%	11.4%
Cybersecurity Analyst	50.9%	46.4%	51.8%	89,429	61.8%	45.5%	11%
UX Designer	43.5%	44.4%	55.6%	90,157	35.2%	48.1%	10.8%
AI Researcher	51.9%	47.1%	54.9%	95,583	35.3%	53.9%	10.2%

Table 9

Classification thresholds for job suitability.

Category	Description	Skill Match Score (%)	Sentiment Score (%)
High Suitability	Strong alignment with user skills and preferences.	80–100	70–100
Moderate Suitability	Partial alignment, indicating some skill gaps.	50–79	50–69
Low Suitability	Minimal alignment, suggesting significant skill mismatches.	0–49	0–49

3.13. Reproducibility

The software can be reproduced if the user follows the same pre-processing procedures, feature extraction methodologies, filtering methods, modeling, metrics, and explanations found in the accompanying tables and figures. All expected inputs, transformations, scoring logic, and outputs are explicitly defined and available to allow for an independent reproduction and validation of the prediction results.

4. Background

Machine learning models have shown strong predictive performance across various domains. Abedi et al. [1] demonstrated the effectiveness of XGBoost, Random Forest, and Boosted Regression Trees for complex prediction tasks. Chicco et al. [2] highlighted the importance of appropriate regression evaluation metrics such as R^2 , RMSE, MAE, MSE, and MAPE for assessing model performance. Recommendation systems have evolved through collaborative filtering and hybrid approaches. Ge et al. [3] proposed collaborative filtering with collective training to improve recommendation accuracy. Hasan et al. [4] applied NLP-based sentiment analysis to social media data, showing the value of textual information in decision-making systems. Ingole et al. [5] developed a machine learning-based job role suggestion system, while Julian and Haripriya [6] introduced an NLP-based resume analysis and adaptive skill assessment framework. Ko et al. [7] provided a comprehensive survey of recommendation techniques and applications. Koren et al. [8] reviewed recent advances in collaborative filtering methods. In the educational domain, Mondal et al. [9] proposed a grade-based course recommendation system, while Naeem et al. [10] compared multiple machine learning algorithms and demonstrated the effectiveness of ensemble models. Patra and Dutta [11] further showed the usefulness of evaluation metrics such as RMSE in predictive modeling. Several studies have focused on skill-oriented recommendation. Qin et al. [12] proposed a skill-oriented question generation and recommendation framework for intelligent interviews. Quan and Raheem [13] explored

employment analytics using specialized skill sets and salary prediction. Sankhe et al. [14] developed a skill-based course recommendation system that identifies skill gaps, while Tan et al. [15] applied recommendation techniques in e-learning environments. Hybrid recommendation methods have also gained attention. Thorat et al. [16] surveyed collaborative, content-based, and hybrid recommendation systems, emphasizing the benefits of combining multiple approaches. In addition, Velioglu et al. [17] demonstrated how Explainable Artificial Intelligence (XAI) can improve model transparency and user trust. The effectiveness of ensemble learning methods was further validated by Zamani Johar-estani et al. [18], who successfully applied Random Forest and XGBoost models for environmental prediction tasks.

Machine learning methods have improved job recommendation systems along various dimensions: predicting job fit against users, predicting required qualifications for jobs, predicting success against different jobs and user profiles, and measuring fit between users and jobs. While many of the models above are collections of individual stand-alone methods, they do not incorporate the many different data sources that can be used to generate recommendations including: skill match between users and jobs, the sentiment associated with job postings, and indicators of the labor market.

The majority of existing job recommendation systems utilize keyword search, content-based filtering, or collaborative filtering (i.e. user preference matches). However, these methods do not generally integrate with predictive modeling, labor market intelligence, or explainability.

To overcome the limitations mentioned above, the proposed system incorporates hybrid recommender systems, machine learning prediction, sentiment analysis, and explainable AI in an integrated framework. The proposed system follows the components of hybrid recommender systems by using both collaborative filtering and content-based filtering to provide more accurate job recommendation solutions amongst a larger pool of available jobs and users.

4.1. Novelty and contribution of the proposed system

The proposed system introduces a hybrid job recommendation framework that integrates multiple components into a unified pipeline. The main contributions are:

- A unified hybrid architecture combining content-based filtering, collaborative filtering, and machine learning prediction in a single workflow.
- Inclusion of labor market-driven features such as AI adoption level, automation risk, and job growth projection.
- A multi-factor weighted scoring mechanism for computing job suitability.

Table 10

Final hybrid job recommendation.

Job Title	Industry	Salary (USD)	Job Growth	Sentiment Score	Job Score	Predicted Score	Skill Match Score
Cybersecurity Analyst	Healthcare	89,856	0.0	0.46	17,971	17,935	3
AI Researcher	Transportation	89,253	0.0	0.27	17,851	17,848	4
Cybersecurity Analyst	Manufacturing	89,024	0.5	0.51	17,805	17,801	3
Cybersecurity Analyst	Energy	88,904	0.5	0.48	17,781	17,784	3
Data Scientist	Technology	88,517	0.5	0.29	17,704	17,702	1

- Integration of Explainable AI (SHAP) to provide transparency in recommendations.
- A context-aware recommendation approach that considers both user profile and evolving job market conditions.

This structured integration of components clearly distinguishes the proposed system from existing approaches.

4.2. Application potential

The recruitment and job matching software market is projected to grow exponentially as online hiring portals, company HR applications, and the growing trend of skills-based hiring become universally adopted. Furthermore, as more and more companies plug-in Digital Recruitment Apps and leverage remote working, the demand for software that accurately matches jobs and assists users with making data-driven career decisions becomes an imperative.

Job Matching Software is already in use on a variety of online job and recruitment sites and via company HR analytics platforms for helping employers find the best candidates for available positions. In addition, Job Matching Software is being developed to evaluate a candidate's skill set, experience, salary expectations, and job growth expansion rate. Future areas of application include Workforce Planning, Skills Gap Analysis, Reskilling, Training Recommendation Systems, and Labor Market Analysis.

The three most established systems in the marketplace, LinkedIn, Indeed and Glassdoor, rely on Keyword Matching, User Activity or Popularity-based ranking to help users find jobs; none of these provide users with reliable or complete transparency and limited support for skill relevance or expected future jobs. Job Matching Software has found its place by distributing the advantages of a Hybrid System, and integrating both Predictive Modeling and Explainable Outputs, providing customers with consistent Job Recommendations that are Transparent and Context Aware; Job Matching Software limitations continue to be largely associated with Data Availability and the need for Periodic Updates.

The inclusion of explainable AI and labor market driven features makes the system more suitable for real world deployment, where transparency and informed decision making are important.

4.3. State of development

The software has been developed as a research prototype and implemented as part of an academic study. Core components, including data processing, feature extraction, filtering, predictive modeling, and explainable outputs, have been fully implemented and evaluated using real-world job data.

The system is not yet a commercial product but is sufficiently mature to support further development, pilot testing, or licensing with additional engineering and deployment work.

Ethics statements

There is the potential for bias to be present in job recommendations due to data distribution, feature selection, and model behavior. For example, if the following were the only features of a candidate, such as only having experience or only salary expectation, those features would have more influence on the recommendation than other features may have; therefore, the potential exists for these features to make recommendation results less equal for candidates of differing genders or ethnicities.

To reduce this risk, this study of a job recommendation system focuses on skill matching and job-related traits to match candidates with jobs. Career affecting demographics (e.g., gender, race/ethnicity, and personal identity) are not a feature of the model; hence it is thought that these features may bias the overall recommendation result.

An added benefit of using explainable AI provides insight into how each of the features in the model impacts the final recommendation. Using this information helps to identify and be aware of any biases or sensitivities that might affect the recommendations.

Currently, fairness is still a significant concern for job recommendation systems, and it is essential to continue to evaluate and monitor the system during the deployment phase to ensure that there are no ongoing biased recommendations.

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CRediT authorship contribution statement

Snehadhanya Dutta: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Moumita Chatterjee:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Dhrubasish Sarkar:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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